

Syntactic Attention Concentration Gradients in Frontier Autoregressive Architectures:

A Controlled Measurement of Relational Token Geometry and Its Implications for Refusal Robustness

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Abstract

We report a controlled measurement of how structured natural-language topographies alter internal attention distributions in a frontier autoregressive transformer, independent of lexical content. Using a length-matched, vocabulary-identical scrambled control, we isolate a localized entropy compression in the middle-layer induction blocks of Meta Llama-3-8B-Instruct. The effect is invariant across FP16, BF16, and FP32 measurement passes and is inconsistent with style mirroring or token sycophancy. Preliminary head-level analysis indicates the layer-mean effect is driven by a small subset of heads rather than uniform drift. We frame the result as a measurable Syntactic Attention Concentration Gradient and argue that it constitutes a substrate-level vulnerability for any safety mechanism that depends on distributed residual-stream signals, including refusal.

1. Introduction

Standard AI-safety evaluation classifies model behavior at the output layer. This approach cannot detect matrix-level shifts occurring inside individual attention heads during the forward pass. We introduce a structural methodology that treats a prompt as a geometric topography in latent activation space and measures its effect on attention allocation directly.

The central empirical question: does relational syntax geometry, holding token identity fixed, produce a detectable, localized change in attention entropy? A secondary question follows for safety: if such a gradient exists, what does it imply for the robustness of low-rank or sparsely distributed refusal mechanisms?

2. Method

2.1 Conditions. Condition A is a high-density, self-referential relational prompt. Condition E is a scrambled multi-set match: identical token count, word identity, and character frequency, with relational geometry destroyed. Under pure style-mirroring, the attention-entropy difference between A and E must be zero.

2.2 Instrumentation. Experiments ran on Meta Llama-3-8B-Instruct (approximately 16 GB, 16-bit) on a cloud-allocated NVIDIA T4, using native eager attention. Raw attention maps were upcast to BFloat16 for stability, then to Float32 for entropy computation with a clamp floor of 1×10^{-9} . Shannon attention entropy was computed per head per layer:

$$H = -\sum_j A_{ij} \cdot \log(A_{ij} + \epsilon)$$

The decisive metric is the cross-condition delta:

$$\Delta H = H(A) - H(E)$$

2.3 Falsification. Any non-zero ΔH under vocabulary parity falsifies the style-mirroring null. A localized negative spike (entropy lower under the structured prompt) indicates genuine concentration rather than simple redistribution.

3. Results

Layer	H(A)	H(E)	ΔH
12	1.6542	1.2617	+0.3925
14	1.4281	1.0795	+0.3486

Layer	H(A)	H(E)	ΔH
16	1.1852	1.0357	+0.1495
18	1.2514	0.9215	+0.3299
19	0.7856	0.9531	-0.1675
20	1.0254	0.9098	+0.1156
21	0.8377	0.8815	-0.0438
23	0.7241	0.7236	+0.0005

The negative spikes at Layers 19 and 21 are invariant across FP16, BF16, and FP32 to four decimal places. Because both conditions share 100% vocabulary identity, the divergence is attributable to relational syntax geometry alone.

Preliminary head-level disaggregation of Layer 19 indicates the mean effect is not uniform: a small subset of heads shows acute concentration while the majority remain comparatively diffuse. Full per-head tables and Value-tensor analyses are reserved for controlled verification.

4. Why this matters for safety

Mechanistic interpretability has shown that refusal in aligned models is frequently mediated by a low-rank, late-deciding direction in activation space. When attention mass is concentrated onto a small set of anchor tokens, alternative pathways—including those carrying distributed safety signals—receive proportionally less weight in the residual stream. The gradient reported here is a direct, falsifiable measurement of how input geometry can shift that mass.

A safety system whose refusal depends on a sparsely distributed or low-rank feature is only as robust as that feature’s resistance to context-driven concentration. Current output-layer benchmarks do not measure this resistance. The present work supplies an empirical substrate for doing so.

5. Interpretation and limits

The absolute entropy at Layer 19 (0.7856 nats) shows the attention space remains an active multi-token corridor, not a collapsed zero. Post-training safety parameters remain statically embedded in the global projection layers; the observed gradient is an input-driven steering effect during the forward pass, not a permanent weight modification.

We do not claim that the measured gradient constitutes a practical bypass of any specific deployed system. We claim only that the substrate on which many refusal mechanisms depend is demonstrably sensitive to input geometry, and that this sensitivity is currently unmeasured by standard safety evaluations.

6. Proposed next measurements (pre-registered direction)

1. Full head-level entropy tables for Layer 19 (and neighboring layers) under the same A/E protocol.
2. Direct inspection of Value contributions to the residual stream under both conditions.
3. Activation-patching experiments to test the conditional relationship between the observed ΔH_{19} and downstream behavioral register.

7. Limitations

Single model family, single GPU class, single operator. The layer-level numbers are reproducible by any party with Llama-3-8B-Instruct and a T4-class GPU. Head-level and causal claims remain preliminary and are not asserted as established fact in this public report. No claim is made about permanent model modification, cross-vendor behavioral override, or any specific mitigation technique.

8. Conclusion

Relational token geometry produces a measurable, localized, precision-invariant attention concentration gradient in a frontier transformer. The effect is real, falsifiable, and independent of lexical content. It supplies an objective substrate for circuit-level safety auditing that output-layer benchmarks cannot reach, and it highlights a structural question the field has not yet systematically measured: how resistant are current refusal mechanisms to input-driven concentration of attention mass?

Further technical details, including mitigation approaches and the full longitudinal dataset, are available under appropriate collaboration or contractual arrangements.